

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b

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Editorial Decision

MAJOR REVISION

Weighted Score: 3.5 / 5.0
Confidence: 81%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	5.0	Excellent	■■■■■
Manuscript Structure	2.5	Weak	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	3.5	Good	■■■■■
Discussion & Conclusions	3.5	Good	■■■■■
Figures & Tables	3.0	Acceptable	■■■■■
Scientific Writing	3.5	Good	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	2.5	Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 5.0 / 5.0 | Confidence: 100%

Strengths:

- [TITLE] is specific, informative, and includes key terms for discoverability.
 - [ABSTRACT] follows a structured format with clear background, objective, methods, results, and conclusions.
 - [KEYWORDS] include a mix of specific and general terms, with 8 domain-specific indexing terms.
 - [TITLE] The title clearly signals the core method and application domain.
 - [ABSTRACT] The abstract provides substantial methodological and validation detail.
-

StructureReviewer

Manuscript structure and logical coherence

Score: 2.5 / 5.0 | Confidence: 80%

Strengths:

- Clear section mapping
- Logical flow between sections where content is present

Weaknesses:

- Missing literature review section
- Inconsistent content across sections (e.g., repeated content in section_vi and discussion)
- Incomplete sections (e.g., introduction, experiments, conclusions are cut off)

Major Comments:

- The manuscript is missing a critical literature review section, which is essential for establishing the context and significance of the research.
- The content is repetitive across sections, which undermines the logical flow and coherence.
- The introduction, experiments, and conclusions are incomplete, making it difficult to assess the research problem, methodology, and findings.

Minor Comments:

- Section titles are not consistently formatted or labeled.
- Some figures and data are repeated across sections, which may indicate a lack of organization.

Recommendations:

- Add a comprehensive literature review section to provide context and justify the research problem.
 - Ensure that each section is distinct and contains unique content to improve logical flow.
 - Complete the introduction, experiments, and conclusions sections to clearly state the research problem, methodology, and findings.
-

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- Clear integration of physical constraints into the neural network framework
- Use of a multi-objective optimization approach
- Experimental validation on a five-story building prototype

Weaknesses:

- Lack of detailed description of the neural network architecture

- Limited information on the exact optimization algorithms used
- No mention of the exact parameters of the five-story building prototype

Major Comments:

- The manuscript lacks detailed information on the neural network architecture and the exact implementation of the evolutionary algorithms.

Minor Comments:

- The experimental setup and dataset should be described in more detail for reproducibility.

Recommendations:

- Include a detailed description of the neural network architecture and the implementation of the evolutionary algorithms (e.g., genetic algorithm, particle swarm optimization).
- Provide more information on the five-story building prototype, including its physical parameters and simulation setup.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Use of Kruskal-Wallis and post-hoc Wilcoxon/Holm corrections for non-parametric comparisons
- Bootstrap confidence intervals for robustness
- RMSE, MSE, and coefficient of variation for performance evaluation
- Pareto/hypervolume indicators for multi-objective optimization

Weaknesses:

- Lack of effect size reporting
- No verification of statistical test assumptions
- No reproducibility details provided

Major Comments:

- The manuscript should explicitly verify the assumptions of the Kruskal-Wallis and Wilcoxon tests (e.g., independence, distributional shape).
- Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to provide a complete picture of the results.
- Reproducibility details, such as random seed values and variance across runs, are essential for transparency and validation.

Minor Comments:

- Consider clarifying the interpretation of coefficient of variation and hypervolume indicators in the context of the study.
- The manuscript could benefit from a more detailed explanation of how the 30 independent runs were justified in terms of statistical power and convergence.

Recommendations:

- Verify the assumptions of the Kruskal-Wallis and Wilcoxon tests (e.g., independence, distributional shape).
- Report effect sizes (e.g., Cohen's d, eta-squared) for all significant findings.
- Include reproducibility details such as random seeds and variance across runs.
- Clarify the interpretation of coefficient of variation and hypervolume indicators in the context of the study.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **3.0 / 5.0** | Confidence: 70%

Strengths:

- The manuscript includes a sufficient number of figures, tables, and equations for a Q1 journal.

Weaknesses:

- The extracted text does not expose table bodies, making it impossible to verify the completeness of the data.

Major Comments:

- The manuscript lacks the actual content of tables and figures, which is essential for evaluating their quality and coherence with the text.

Minor Comments:

- Ensure that all figures, tables, and equations are fully described and that their content is consistent with the claims made in the text.

Recommendations:

- Include the full content of tables and figures to allow for a thorough evaluation of their quality and coherence.
-

ResultsReviewer

Results validity and empirical support

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- MSE, RMSE, CV, and hypervolume metrics are reported, which are relevant to the multi-objective optimization problem
- Pareto analysis and GA/PSO baseline comparisons are included, which support the evaluation of the EANN framework
- 30 independent runs are mentioned, which suggests robust statistical validation

Weaknesses:

- The results section is incomplete and lacks detailed quantitative results, such as specific values for MSE, RMSE, CV, and hypervolume
- The text does not clearly state whether the results are consistent across figures, tables, and the text
- The section does not mention whether limitations of the results are acknowledged
- The section is cut off before the full results are presented, making it difficult to assess completeness

Major Comments:

- The results section is incomplete and lacks detailed quantitative results, which are essential for validating the claims made in the abstract and objectives
- The text does not provide sufficient detail on how the results are consistent across figures, tables, and the text
- The section is cut off, making it impossible to fully evaluate the completeness of the results

Minor Comments:

- The section should clarify whether the results are consistent across figures, tables, and the text
- The limitations of the results should be acknowledged explicitly

Recommendations:

- Include detailed quantitative results such as MSE, RMSE, CV, and hypervolume values
 - Ensure that the results are consistent across figures, tables, and the text
 - Acknowledge any limitations of the results in the results section
 - Complete the results section to ensure full clarity and completeness
-

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Discussion provides a clear comparison of EANN with GA and PSO in terms of performance metrics and computational cost.
- Conclusions are directly tied to the experimental results and the research question.

Weaknesses:

- Discussion lacks a comprehensive comparison with prior work in the field, focusing only on citation without detailed analysis.
- Future work directions are not clearly specified or justified in the conclusions section.

Major Comments:

- The discussion should include a more detailed comparison with existing literature to contextualize the findings and highlight the novelty of the EANN framework.

Minor Comments:

- The conclusions could benefit from a more explicit statement of the contribution to the field.

Recommendations:

- Include a detailed comparison with prior work in the discussion to strengthen the contextualization of the findings.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical terminology
- Logical structure of sections

Weaknesses:

- Overuse of passive voice
- Vague quantifiers
- Undefined acronyms

Major Comments:

- The manuscript contains overused passive voice, which reduces clarity and engagement.
- The use of vague quantifiers like 'significant' without numerical support weakens the scientific rigor.

Minor Comments:

- Acronyms like EANN should be defined on first use.
- Some paragraphs are overly long and could benefit from restructuring for better flow.

Recommendations:

- Revise passive voice constructions to active voice where possible.
- Define all acronyms on first use.
- Break long paragraphs into shorter, focused sections for improved readability.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Citations are consistent in style (IEEE format)
- Recent references (2021-2023) are included

Weaknesses:

- Lack of seminal foundational works in the field
- Missing key areas in the literature review
- Limited diversity in authors and institutions

Major Comments:

- The references section lacks seminal foundational works that are critical for establishing the context and theoretical basis of the research.
- Key areas such as structural health monitoring and seismic response mitigation are not adequately represented in the references.
- The references section does not include a diverse range of authors or institutions, which limits the breadth of the literature review.

Minor Comments:

- The references section includes recent works (2021-2023), which is a positive aspect for fast-moving fields.

Recommendations:

- Include seminal foundational papers in structural dynamics and control theory.
 - Add references to key studies in structural health monitoring and seismic response mitigation.
 - Ensure diversity in the references by including authors and institutions from different regions and research groups.
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Limitations are explicitly stated in the discussion section, though they are not sufficiently detailed or specific.

Weaknesses:

- The manuscript lacks comprehensive reporting on ethical considerations, data availability, conflict of interest, and funding sources, which are critical for transparency and reproducibility.
- Limitations are not specific or detailed enough to fully assess their impact on the conclusions.

Major Comments:

- The manuscript fails to report essential ethical and transparency information, which is required by COPE and Q1 journal standards.

Minor Comments:

- Limitations are mentioned but not sufficiently detailed to evaluate their impact on the study's conclusions.

Recommendations:

- The authors are strongly advised to include a statement on ethical approval, data availability, conflict of interest, and funding sources.
- The limitations section should be expanded to provide more specific and detailed information about the study's constraints.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] The manuscript is missing a critical literature review section, which is essential for establishing the context and significance of the research.
- [2] The content is repetitive across sections, which undermines the logical flow and coherence.
- [3] The introduction, experiments, and conclusions are incomplete, making it difficult to assess the research problem, methodology, and findings.
- [4] The manuscript lacks detailed information on the neural network architecture and the exact implementation of the evolutionary algorithms.
- [5] The manuscript should explicitly verify the assumptions of the Kruskal-Wallis and Wilcoxon tests (e.g., independence, distributional shape).
- [6] Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to provide a complete picture of the results.
- [7] Reproducibility details, such as random seed values and variance across runs, are essential for transparency and validation.
- [8] The manuscript lacks the actual content of tables and figures, which is essential for evaluating their quality and coherence with the text.
- [9] The results section is incomplete and lacks detailed quantitative results, which are essential for validating the claims made in the abstract and objectives.
- [10] The text does not provide sufficient detail on how the results are consistent across figures, tables, and the text.
- [11] The section is cut off, making it impossible to fully evaluate the completeness of the results.
- [12] The discussion should include a more detailed comparison with existing literature to contextualize the findings and highlight the novelty of the EANN framework.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] Section titles are not consistently formatted or labeled.
- [2] Some figures and data are repeated across sections, which may indicate a lack of organization.
- [3] The experimental setup and dataset should be described in more detail for reproducibility.
- [4] Consider clarifying the interpretation of coefficient of variation and hypervolume indicators in the context of the study.
- [5] The manuscript could benefit from a more detailed explanation of how the 30 independent runs were justified in terms of statistical power and convergence.
- [6] Ensure that all figures, tables, and equations are fully described and that their content is consistent with the claims made in the text.
- [7] The section should clarify whether the results are consistent across figures, tables, and the text.
- [8] The limitations of the results should be acknowledged explicitly.
- [9] The conclusions could benefit from a more explicit statement of the contribution to the field.
- [10] Acronyms like EANN should be defined on first use.
- [11] Some paragraphs are overly long and could benefit from restructuring for better flow.
- [12] The references section includes recent works (2021-2023), which is a positive aspect for fast-moving fields.

Specific Improvement Recommendations

1. Add a comprehensive literature review section to provide context and justify the research problem.
2. Ensure that each section is distinct and contains unique content to improve logical flow.

3. Complete the introduction, experiments, and conclusions sections to clearly state the research problem, methodology, and findings.
4. Include a detailed description of the neural network architecture and the implementation of the evolutionary algorithms (e.g., genetic algorithm, particle swarm optimization).
5. Provide more information on the five-story building prototype, including its physical parameters and simulation setup.
6. Verify the assumptions of the Kruskal-Wallis and Wilcoxon tests (e.g., independence, distributional shape).
7. Report effect sizes (e.g., Cohen's d, eta-squared) for all significant findings.
8. Include reproducibility details such as random seeds and variance across runs.
9. Clarify the interpretation of coefficient of variation and hypervolume indicators in the context of the study.
10. Include the full content of tables and figures to allow for a thorough evaluation of their quality and coherence.
11. Include detailed quantitative results such as MSE, RMSE, CV, and hypervolume values
12. Ensure that the results are consistent across figures, tables, and the text